The equation $\sigma(n) = 2\sigma^*(n)$: the single-repeated-prime case and the structure of hypothetical exceptions

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Abstract

Let $\sigma^*(n)$ denote the sum of unitary divisors of n. OEIS sequence A063880 lists the solutions of $\sigma(n) = 2\sigma^*(n)$; a conjecture recorded by R. Stephan (2003) asserts that every solution is congruent to 108 (mod 216). We prove the conjecture unconditionally for all solutions having at most one repeated odd prime factor: such solutions are exactly $n = 2^2 \cdot 3^3 \cdot s$ with s squarefree and $\gcd(s, 6) = 1$. This class contains every known solution (all 56,297 solutions below 2×10^7). We further prove that no solution has exactly two repeated odd prime factors together with $v_2(n) \leq 2$, and we give quantitative structure theorems confining any further hypothetical solutions. The proofs are elementary, resting on a 2-adic valuation identity obtained by lifting the exponent and on interval arithmetic for the local ratios $\sigma(p^a)/(p^a + 1)$. We locate precisely the residual obstruction, contrast the situation with the pure-unitary sibling problem, and state falsifiable expectations for the remaining cases.

1 Introduction

For $n = \prod_p p^{a_p}$ let $\sigma(n) = \prod_p (1 + p + \cdots + p^{a_p})$ be the sum of divisors and $\sigma^*(n) = \prod_p (p^{a_p} + 1)$ the sum of *unitary* divisors (divisors $d \mid n$ with $\gcd(d, n/d) = 1$). This paper concerns the solutions of

$$\sigma(n) \;=\; 2\,\sigma^*(n), \tag{1}$$

tabulated as OEIS sequence A063880 [1]. The sequence begins $108, 540, 756, 1188, \ldots$, and two conjectures are recorded in the entry: that every solution satisfies $n \equiv 108 \pmod{216}$ (R. Stephan, 2003), and that 108 is the unique primitive solution. A. Eldar (2024) reduced the congruence conjecture to a statement about the repeated prime factors of solutions; the present paper is organized around that reduction. Since $n \equiv 108 \pmod{216}$ is equivalent to $v_2(n) = 2$ and $27 \mid n$, the conjecture is a rigidity statement: the 2- and 3-adic structure of every solution is completely determined.

Our main results are the following. Call an odd prime p *repeated* in n if $p^2 \mid n$.

[*]This paper was researched and written by an AI system (Anthropic's Claude Fable 5) across interactive sessions on July 20, 2026; see Section 8 for a full provenance, verification, and contribution statement.

[†]E.K.'s contribution was the selection of the problem and the research direction, including the independent-verification workflow; the mathematical content, proofs, and computations are due to the first author. This division of labor is stated at E.K.'s request.

Theorem 1. *The solutions of* (1) *with at most one repeated odd prime are exactly*

$$n \;=\; 2^2 \cdot 3^3 \cdot s, \qquad s \text{ squarefree}, \ \gcd(s,6)=1.$$

In particular every such solution satisfies $n \equiv 108 \pmod{216}$.

Theorem 2. *No solution of* (1) *has exactly two repeated odd primes together with* $v_2(n) \le 2$.

Theorem 1 proves Stephan's conjecture on a class containing every known solution: an exhaustive computation shows that all 56,297 solutions below 2×10^7 have exactly one repeated odd prime, and indeed that the set of solutions below that bound coincides exactly with $\{108s : s \text{ squarefree}, \ \gcd(s,6)=1\}$. Theorem 2 then closes the nearest hypothetical exception class. Combining the two with elementary structure theorems (Section 6) yields the corrected logical map:

Stephan's conjecture holds if and only if no solution has exactly two repeated odd primes and every solution with three or more repeated odd primes has $v_2(n)=2$ *and* $27 \mid n$.

Context and related work

Equation (1) is a mixed relative of two well-studied families. The pure-unitary perfect equation $\sigma^*(n)=2n$ (Subbarao–Warren) was recently subjected to a systematic bounded-box reduction by Maciejewski [2], whose paper is also the model for the shape of our residual analysis: prove the tractable part unconditionally, verify a computational frontier, and locate the analytic obstruction precisely. Yamada [3] gives a complete elementary resolution of the cousin equation $\sigma^*(\sigma^*(N))=2N$ in the odd case, demonstrating what a full elementary endgame in this genre looks like; the nearest classical study of iterated/mixed unitary equations is Sitaramaiah–Subbarao [4].

A structural contrast is worth stating explicitly, because it explains why Theorem 1 is short while the analogous single-prime frontier of the pure-unitary problem is not. In $\sigma^*(n)=2n$, every prime of the support is forced back into the support through the factors p^e+1, producing the self-referential dependency cascade that Maciejewski's filter apparatus is built to control. In (1) that closure is absent: the numerators $\sigma(p^a)$ may contain primes foreign to n, and the equation constrains only the global product of local ratios. This looser coupling removes the cascade — and in exchange admits an exact 2-adic identity (Lemma 1) with no analogue in the pure-unitary setting. The single-repeated-prime case then closes by a two-sided interval squeeze and one congruence.

2 Preliminaries

Throughout, $n=2^e m$ with m odd, $m=\prod_{p \text{ odd}} p^{a_p}$, and

$$\rho(p^a) \;=\; \frac{\sigma(p^a)}{p^a+1}, \qquad D \;=\; 2^{e+1}-1.$$

Since σ and σ^* are multiplicative, (1) is equivalent to $\prod_{q \,\|\, n} \rho(q)=2$ over the maximal prime powers q of n. A direct computation gives $2/\rho(2^e)=2(2^e+1)/(2^{e+1}-1)=1+3/D$, so (1) is equivalent to

$$\prod_{p \mid m} \rho(p^{a_p}) \;=\; 1+\frac{3}{D}. \tag{2}$$

For $e = 1$ note $\sigma(2m) = 3\sigma(m)$ and $\sigma^*(2m) = 3\sigma^*(m)$, so the cases $e \in \{0, 1\}$ both reduce to $\sigma(m) = 2\sigma^*(m)$ on the odd part, i.e. to (2) with right-hand side 2.

We use the following elementary facts about ρ, each a one-line verification:

(i) $\rho(p) = 1$; only repeated primes contribute to (2).

(ii) $\rho(p^a)$ is strictly increasing in a, with $\rho(p^a) < \dfrac{p}{p-1}$ for all a.

(iii) $\rho(p^3) = \dfrac{1+p^2}{p^2-p+1}$.

(iv) $\rho(p^a) - 1 > \dfrac{1}{p+1}$ for $a \geq 2$.

3 A 2-adic identity

Lemma 1. *If $\sigma(n) = 2\sigma^*(n)$, then, summing over the odd primes dividing n,*

$$\sum_{a_p \text{ odd}} \big(v_2(a_p+1) - 1\big) \;=\; 1 \;+\; \#\{p \text{ odd} : a_p \text{ even}\}.$$

Proof. Compare v_2 of both sides multiplicatively. The factors at $p = 2$, namely $\sigma(2^e) = 2^{e+1} - 1$ and $2^e + 1$, are odd. For odd p with a even, $\sigma(p^a)$ is a sum of an odd number of odd terms, hence odd, while $p^a \equiv 1 \pmod 8$ gives $v_2(p^a + 1) = 1$. For odd p with a odd, the cofactor of $p + 1$ in $p^a + 1$ is an alternating sum of a odd terms, hence odd, so $v_2(p^a + 1) = v_2(p + 1)$; and by lifting the exponent, $v_2(p^{a+1} - 1) = v_2(p - 1) + v_2(p + 1) + v_2(a + 1) - 1$, whence $v_2(\sigma(p^a)) = v_2(p + 1) + v_2(a + 1) - 1$. Equating $v_2(\sigma(n)) = 1 + v_2(\sigma^*(n))$ and cancelling $\sum v_2(p + 1)$ over odd exponents gives the identity. □

Corollary 1. *Every solution has an odd prime with exponent $\equiv 3 \pmod 4$; in particular $p^3 \mid n$ for some odd p, and the odd part of a solution is never squarefree.*

Proof. The right-hand side of the identity is at least 1, so some term on the left is positive, forcing $v_2(a_p + 1) \geq 2$, i.e. $a_p \equiv 3 \pmod 4$. □

4 Proof of Theorem 1

Suppose n is a solution with at most one repeated odd prime. By Corollary 1 there is exactly one, say p with exponent $a \geq 2$. If a were even, the right side of Lemma 1 would be 2 against a vanishing left side; so a is odd, and the identity forces $v_2(a + 1) = 2$, hence $a \equiv 3 \pmod 8$ and $a \geq 3$. All other odd primes have exponent 1 and contribute $\rho = 1$, so (2) reads

$$\rho(p^a) \;=\; 1 + \frac{3}{D}. \tag{3}$$

Case $e \in \{0, 1\}$. The right side equals 2, while $\rho(p^a) < p/(p - 1) \leq 3/2$. Impossible.

Case $e \geq 2$. From fact (ii), $p/(p - 1) > 1 + 3/D$ gives $3(p - 1) < D$. From facts (ii)–(iii) and $a \geq 3$,

$$1 + \frac{3}{D} \;=\; \rho(p^a) \;\geq\; \rho(p^3) = \frac{1+p^2}{p^2-p+1} \quad\Longrightarrow\quad D \leq 3p - 3 + \frac{3}{p} < 3p.$$

Hence $D < 3p < D+3$, i.e. $3p \in \{D+1, D+2\}$. Now clear denominators in (3):

$$D\,\sigma(p^a) \;=\; (D+3)\,(p^a+1).$$

Reducing modulo p, both $\sigma(p^a)$ and p^a+1 are $\equiv 1$, so $D \equiv D+3 \pmod{p}$ and $p \mid 3$. Thus $p=3$ and $3p=9$ forces $D \in \{7,8\}$; D is odd, so $D=7$ and $e=2$. Finally $7\,\sigma(3^a) = 10\,(3^a+1)$ simplifies to $3^a = 27$, so $a=3$.

Conversely, for $n = 2^2 \cdot 3^3 \cdot s$ with s squarefree and coprime to 6, $\rho(4)\rho(27) = \frac{7}{5}\cdot\frac{10}{7} = 2$ and every other ρ-factor is 1, so n solves (1); and $n = 108s$ with s odd gives $n \equiv 108 \pmod{216}$. □

5 Proof of Theorem 2

Let $p < q$ be the two repeated odd primes, with exponents $a, b \geq 2$.

Case $e \in \{0,1\}$. Equation (2) has right side 2, but by fact (ii) $\rho(p^a)\rho(q^b) < \frac{3}{2}\cdot\frac{5}{4} = \frac{15}{8} < 2$. Impossible.

Case $e=2$. Now $\rho(p^a)\rho(q^b) = \frac{10}{7}$. Since $q > p$, both factors are $< p/(p-1)$, so $\big(p/(p-1)\big)^2 > \frac{10}{7}$, i.e. $3p^2 - 20p + 10 < 0$, giving $p \in \{3,5\}$.

Subcase $p=3$. Since $\rho(3^3) = \frac{10}{7}$ exactly (which would force $\rho(q^b)=1$, impossible) and $\rho(3^a) > \frac{10}{7}$ for $a \geq 4$, we must have $a=2$ and

$$\rho(q^b) \;=\; T := \frac{10/7}{13/10} = \frac{100}{91}.$$

From $\rho(q^b) < q/(q-1)$ we get $q < 1 + (T-1)^{-1} = \frac{100}{9} < 12$, so $q \in \{5,7,11\}$. For $q \in \{5,7\}$ already $\rho(q^2) > T$ ($\frac{31}{26}, \frac{57}{50} > \frac{100}{91}$). For $q=11$, exact cross-multiplication gives

$$\rho(11^2) = \tfrac{133}{122} < \tfrac{100}{91} < \tfrac{122}{111} = \rho(11^3) \qquad (133\cdot 91 = 12103 < 12200, \quad 122 \cdot 91 = 11102 > 11100),$$

and $\rho(11^b)$ is increasing, so no b attains T.

Subcase $p=5$. Since $\rho(5^a) < \frac{5}{4}$, we need $\rho(q^b) > \frac{10}{7}\cdot\frac{4}{5} = \frac{8}{7}$, so $q/(q-1) > \frac{8}{7}$, so $q < 8$, so $q=7$. If $b=2$: $\rho(7^2) = \frac{57}{50} < \frac{8}{7}$, impossible. If $b \geq 3$: $\rho(7^b) \geq \rho(7^3) = \frac{50}{43}$; but $a=2$ would require $\rho(7^b) = \frac{10/7}{31/26} = \frac{260}{217} > \frac{7}{6}$, exceeding every value of $\rho(7^b)$, while $a \geq 3$ would require $\rho(7^b) = \frac{10/7}{\rho(5^a)} \leq \frac{10}{7}\cdot\frac{21}{26} = \frac{15}{13} < \frac{50}{43}$. No case survives. □

Remark 1. The escape at $q=11$ is by a margin of $\frac{100}{91} - \frac{133}{122} = \frac{2\cdot 97}{11102}$ on one side and $\frac{122}{111} - \frac{100}{91} = \frac{2}{10101}$ on the other: the interval arithmetic in Theorem 2 is sharp, not slack. An independent brute force over $p < 100$, $q < 500$, exponents up to 30, in exact rational arithmetic, confirms emptiness.

Proposition 1 (Parity classes)**.** *A solution with exactly two repeated odd primes cannot have both exponents even, and the exponent pair lies in one of two classes: both odd with* $\{v_2(a+1), v_2(b+1)\} = \{1,2\}$ *(one exponent* $\equiv 1 \bmod 4$*, hence* ≥ 5*; the other* $\equiv 3 \bmod 8$*), or one exponent even and the other* $\equiv 7 \pmod{16}$*, hence* ≥ 7*.*

Proof. Immediate from Lemma 1 with the right side equal to 1, 2 respectively. □

6 The residual case

By Theorem 2, any hypothetical solution with exactly two repeated primes has $v_2(n) \geq 3$ and therefore already violates Stephan's conjecture (which requires $v_2(n) = 2$). Solutions with $k \geq 3$ repeated primes could in principle satisfy the congruence, provided $v_2 = 2$ and $27 \mid n$. Hence:

Proposition 2 (Logical map)**.** *Stephan's conjecture is equivalent to the conjunction:* (a) *no solution has exactly two repeated odd primes, and* (b) *every solution with at least three repeated odd primes has $v_2(n) = 2$ and $v_3(n) \geq 3$.*

Quantitative structure confines the hypotheticals. For any solution and any repeated prime p, fact (iv) and $\rho(p^a) - 1 \leq \prod \rho - 1 = 3/D$ give

$$\frac{1}{p+1} \;<\; \rho(p^a) - 1 \;\leq\; \frac{3}{D}, \qquad \text{hence} \qquad p \;>\; \frac{2^{e+1}-4}{3}. \tag{4}$$

For $k = 2$ and $e \geq 3$ the upper-window bound $\big(p/(p-1)\big)^2 > 1 + 3/D$ confines the smaller prime below roughly $2D/3$, so both repeated primes lie in a window of width $\approx D/3$ around $D/2$, with exponents in the classes of Proposition 1; any such solution exceeds roughly 2^{6e}. For $e \in \{0, 1\}$ (right side 2 in (2)), at least three repeated primes are forced, and at least seven if 3 is not among them, since $\frac{5}{4} \cdot \frac{7}{6} \cdot \frac{11}{10} \cdot \frac{13}{12} \cdot \frac{17}{16} \cdot \frac{19}{18} < 2$.

Outlook

We record falsifiable expectations. The $k = 2$, $e \geq 3$ flank appears closable by the methods of this paper: in the window (4) the deviations $\rho - 1$ differ from $1/(p-1)$ by terms of size $\approx 2^{-4e}$, small enough that clearing denominators should quantize the equation to a bounded integer family, each member dying by the congruence trick of Section 4; the low-exponent branches of Proposition 1 require separate but similarly bounded treatment. By contrast, the chamber $e \in \{0, 1, 2\}$ with $k \geq 3$ permits small repeated primes with exponents accumulating at $p/(p-1)$ — the same obstruction that Maciejewski [2] identifies for the sibling problem, and the point at which elementary methods in this genre historically stall. We know of no mechanism (no 3-adic analogue of Lemma 1) that would deliver $27 \mid n$ for such solutions short of proving their nonexistence. A realistic terminal state for elementary methods is therefore Proposition 2 with clause (a) fully proved and clause (b) open on a single precisely-bounded class.

7 Computational verification

All claims were verified in exact arithmetic. Sieving σ and σ^* to $N = 2 \times 10^7$: the solution set below N has 56,297 elements and *equals* $\{108s : s \text{ squarefree},\ \gcd(s, 6) = 1,\ 108s < N\}$ as a set; Lemma 1 and Corollary 1 hold on every element; every element has $v_2 = 2$ and exponent pattern $(3, 1, \ldots, 1)$ on its odd part. Brute force of $D\,\sigma(p^a) = (D+3)(p^a+1)$ over $1 \leq e \leq 25$, $p < 5000$, $2 \leq a < 80$, and of $\sigma(p^a) = 2(p^a+1)$, finds only $(e, p, a) = (2, 3, 3)$. The finite checks in Theorem 2 were performed in exact rational arithmetic, and an independent grid search (Remark 1) finds no solutions. Side computations in the same sessions screened the open conjectures in OEIS A067720 (to $k = 2 \times 10^5$) and A056777 (to 2×10^7) without counterexamples.

8 Provenance, verification status, and contributions

This paper was produced by an AI system (Claude Fable 5, Anthropic) in interactive sessions on July 20, 2026, with E.K. directing the research program; the session transcripts constitute a complete and reproducible record of the derivations, the computations, and their outputs, including one error (an equivalence overclaimed in an early draft of Proposition 2) caught and corrected within the same process. An independent literature review by a second AI system (Claude Opus) verified: that Theorem 1 is not contained in or trivially implied by [2], whose filter apparatus requires a support-closure property absent from equation (1); that the reduction framing follows Eldar (2024) and parallels [2]; and the reachability of the references below. Two items remain open on the novelty side and are flagged rather than claimed: whether Lemma 1 appears in some form in [4], which could not be accessed; and Theorem 2, which postdates the literature review and awaits the same scrutiny. Readers should weight the results accordingly: the mathematics is complete and machine-checked at every numerical step, while the novelty claims are conservative and partially unverified.

References

[1] OEIS Foundation Inc., *Sequence A063880: numbers n such that $\sigma(n) = 2\,\mathrm{usigma}(n)$*, The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A063880`. Comments by R. Stephan (2003) and A. Eldar (2024). See also A348506.

[2] A. Maciejewski, *Bounded-box reductions in the Subbarao–Warren problem for unitary perfect numbers*, preprint (2026), `https://arxiv.org/abs/2605.20475`.

[3] T. Yamada, *Unitary super perfect numbers*, preprint (2008), `https://arxiv.org/abs/0802.4377`.

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[5] Y. Cao, R. Guo, H. Qiu, Z. Feng, and W. Gao, *Every nonnegative integer is a sum of a triangular, a pentagonal, and a heptagonal number*, preprint (2026), `https://arxiv.org/abs/2606.26035`. (Cited as an instance of machine-assisted resolution of a conjecture of the same OEIS-curated genre.)